
DIFFUSION MODELS FOR PREDICTING FUNCTIONAL MOLECULE CONFORMERS

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ABSTRACT

Predicting accurate three-dimensional conformers of functional molecules is a fundamental task in computational chemistry and drug design. While traditional methods rely on molecular mechanics or ab initio sampling, these approaches can be slow and computationally intensive. In this project, we explore the recently proposed All-atom Diffusion Transformer (ADiT), a two-stage latent diffusion model that leverages a variational autoencoder (VAE) and a denoising transformer trained using flow-matching objectives. We extend ADiT by incorporating energy-aware fine-tuning to improve the quality of generated conformers. Through detailed experiments and ablation studies on the QM9 dataset, we show that ADiT can learn chemically valid latent representations and generate high-quality molecular conformers with competitive accuracy and diversity, with minimal supervision.

1 INTRODUCTION

Molecular *conformers* are distinct three-dimensional arrangements of a molecule that arise from rotations about single bonds. Each conformer occupies a particular minimum (or low-lying transition state) on the global potential-energy surface, and the resulting geometrical differences can translate into markedly different physicochemical and biological behavior. For instance, a ligand’s binding affinity, a catalytic transition state, or the packing motif of a molecular crystal can be dominated by whichever conformers are thermally accessible. Conventional sampling pipelines rely on Molecular Dynamics (MD) or Quantum Mechanical (QM) methods such as Density Functional Theory (DFT) to enumerate low-energy states. While these physics based approaches are accurate, their cost scales as $\mathcal{O}(N^3 - N^4)$ with system size and grows linearly with the number of candidate conformers, so an exhaustive DFT search for a single drug-like molecule can take *days* on a multi-core workstation. This cost barrier rules out systematic screening of the millions of molecules needed for modern drug discovery, catalysis, and emerging materials-science workflows.

Generative modeling offers an attractive alternative by learning the *distribution* of chemically valid conformers and sampling from it in milliseconds. Variational autoencoders (VAEs) that embed molecular graphs or 3-D geometries into a smooth latent space (Mansimov et al., 2019; Axelrod & Gomez-Bombarelli, 2022), invertible normalizing flows that map between cartesian coordinates and latent variables (Xu & Dror, 2022), and score-based diffusion models that iteratively denoise structures (Hoogeboom et al., 2022; Xu et al., 2023; Daigavane et al., 2024) have each demonstrated 10^2 – $10^3\times$ speed-ups over MD/ DFT while maintaining high geometric validity. Many of these methods build on rotationally and translationally equivariant neural networks. Some examples are EGNNs or SE(3) Transformers, which can encode the fundamental symmetries of molecular systems. Yet most existing generative frameworks treat *energy* only implicitly (if at all), so they can produce geometrically plausible but thermodynamically unrealistic conformers.

The recently proposed **All-atom Diffusion Transformer** (ADiT) moves the field forward through a two-stage pipeline: **(i)** an all-atom VAE learns a compact latent representation of sets of 3-D coordinates, and **(ii)** a Diffusion Transformer (DiT) trained via flow matching generates novel latents, which are then decoded back to full atomic detail. ADiT avoids hand-crafted torsion features, enforces permutation and Euclidean symmetries, and has achieved state-of-the-art geometric validity across molecules and crystals. Its key limitation, however, is the absence of an energetic inductive bias, allowing high-energetic structures to slip through. We address this gap with an *energy-aware* extension to ADiT: a lightweight MLP head predicts the 0 K internal energy from the pooled latent, and a scalar penalty term augments the flow-matching loss to align generated conformers with reference QM energies. As we show later, this hybrid objective narrows the mean absolute energy error while preserving the diversity and validity benefits of the original model, bridging the divide between data-driven generation and physics-based accuracy.

2 BACKGROUND

2.1 VARIATIONAL AUTOENCODER

Variational autoencoders (VAEs) (Kingma & Welling, 2014) cast generative modelling as Bayesian inference: an encoder $q_\phi(z | x)$ maps data x to a latent distribution and a decoder $p_\theta(x | z)$ reconstructs the input from sampled latents z . For *molecular* and *crystalline* structures the data tuple

$$x = (\mathbf{A}, \mathbf{R}, \mathbf{F}, \mathbf{L})$$

contains atom types $\mathbf{A}$, Cartesian coordinates $\mathbf{R}$, fractional coordinates $\mathbf{F}$, and (for periodic systems) lattice parameters $\mathbf{L}$. Modern encoders treat the atom set as an order-invariant graph: message-passing networks, equivariant transformers, or standard transformers trained with random roto-translations. The decoder mirrors this structure, outputting distributions over atom identities (categorical cross-entropy), coordinates (isotropic MSE), and cell vectors. A weighted reconstruction loss $\mathcal{L}_{\text{rec}}$ plus a KL term $\beta \text{D}_{\text{KL}}[q_\phi(z | x) \parallel \mathcal{N}(0, I)]$ encourages a smooth, approximately Gaussian latent space that is simultaneously expressive and generative. Because the latent set preserves permutation symmetry and (optionally) SE(3) equivariance, it provides a compact yet faithful representation for downstream generative models.

2.2 DENOISING TRANSFORMER

Diffusion Transformers (DiT) (Peebles & Xie, 2023) inject diffusion-time conditioning into every self-attention block of a vanilla transformer, yielding a powerful sequence model that predicts a clean signal from a noisy one. Given a clean sample x_0 , a forward *variance-preserving* process adds Gaussian noise to obtain $x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$. The network is trained under variants of the mean-squared error

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{x_0, t, \epsilon} [\|f_\theta(x_t, t) - \epsilon\|^2]$$

(noise prediction) or an equivalent x_0 / velocity parameterization. While self-attention grants global context, score-matching objectives can demand hundreds of denoising steps at sampling time and expose the model to high-variance gradients—issues that motivate alternative continuous-time formulations.

2.3 FLOW-MATCHING

Flow-matching (FM) (Lipman et al., 2023) recasts diffusion as learning a vector field that deterministically transports points from an easy base distribution to the data manifold. One draws a pair (x_0, x_1) with $x_1 \sim \mathcal{N}(0, I)$, samples an intermediate time $t \sim \mathcal{U}[0, 1]$, forms the interpolation $x_t = (1 - t)x_1 + tx_0$, and supervises the model on the ground-truth velocity

$$u_t(x_t) = \frac{x_0 - x_1}{1 - t}.$$

The loss

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0, x_1, t} [\|f_\theta(x_t, t) - u_t(x_t)\|^2]$$

is fully supervised, avoids score estimation, and empirically converges faster while enabling fewer numerical function evaluations (NFE) during generation.

2.4 ALL-ATOM DIFFUSION TRANSFORMER

Architecture synergy. The All-atom Diffusion Transformer (ADiT) (Joshi et al., 2025) fuses the three ingredients above into a single foundation model for 3-D generative chemistry. Stage I trains an all-atom VAE whose latent $set Z = \{z_i\}_{i=1}^N$ encodes both periodic and non-periodic structures in a common, permutation-equivariant space. Stage II freezes this encoder and learns a flow-matching DiT directly in latent space, drastically reducing dimensionality and circumventing the mixed categorical–continuous manifolds tackled by earlier equivariant score-based models (Hoogeboom et al., 2022; Xu et al., 2023).

Performance rationale. Working in low-dimensional latents means (i) **efficiency**: DiT needs far fewer parameters and integration steps, (ii) **transferability**: the same second-stage network handles small organic molecules, metal-organic frameworks, and inorganic crystals, and (iii) **validity**: chemical constraints learned by the VAE carry through to the generative process. Empirically, ADiT achieves state-of-the-art validity and diversity on QM9 molecules and MP20 crystals while sampling one to two orders of magnitude faster than equivariant score-based baselines—establishing a practical path toward unified, large-scale 3-D generative modelling.

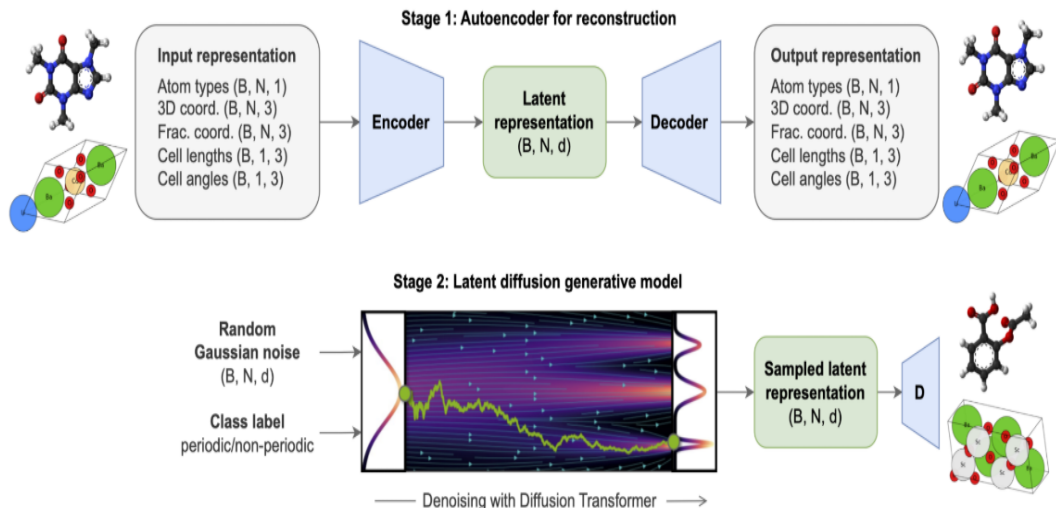


Figure 1: Schematic workflow of the All-atom diffusion transformer from the original paper Joshi et al. (2025)

3 METHODS

3.1 DATA PREPROCESSING

We work with the QM9 dataset, which provides equilibrium 3D geometries and twelve quantum-chemical properties for 133,885 charge-neutral organic molecules composed of H, C, N, O, and F atoms. We use an 80%, 10%, 10% train, valid, and test split. The data is directly available via PyTorch Geometric, and each Data object stores (i) atomic numbers, (ii) Cartesian coordinates, (iii) the 0 K energy, (iv) the molecule size, and (v) a dataset-index label. For the work shown here, we do not incorporate the periodic structures and MOFs in the generation process.

3.2 MODEL ARCHITECTURES

Stage 1: Variational Autoencoder. Both encoder and decoder are implemented with SE(3)-equivariant Transformers (Equiformer-V2 blocks) so that the learned latent codes respect roto-translational symmetry. The encoder converts each atom i with type a_i and position $\mathbf{r}_i$ into

a latent vector $\mathbf{z}_i \in \mathbb{R}^D$; stacking over the N atoms in a mini-batch of size B yields the tensor $\mathbf{X}_1 \in \mathbb{R}^{B \times N \times D}$ with $D = 8$ in all experiments.

Stage 2: Diffusion Transformer (DiT). The denoiser takes the latent tensor $\mathbf{X}_t = (1 - t)\mathbf{X}_0 + t\mathbf{X}_1$ produced by the linear interpolant and predicts the clean sample $\hat{\mathbf{X}}_1$. Its architecture mirrors the text-to-image DiT (Peebles & Xie, 2023) but operates on per-atom latent tokens:

- **Embedding size:** $d_x = D = 8$ (matches the VAE latent channel).
- **Transformer width:** $d_{\text{model}} = 768$.
- **Heads / Layers:** $n_{\text{head}} = 12$, $n_{\text{layer}} = 12$.
- **Self-conditioning:** 50% probability (frozen copy of the previous prediction concatenated to the current input).
- **Class tokens:** a two-way embedding encodes whether the sample is periodic or non-periodic (`num_datasets = 2`).

3.3 ENERGY-AWARE FINE-TUNING

To incorporate energetic priors during fine-tuning, we augment the latent diffusion transformer with a lightweight energy prediction head. After the denoiser produces a latent trajectory, we pool the masked latent tokens of the final clean sample with a mean-reduction and feed the resulting D -dimensional vector to a two-layer MLP that outputs a single scalar $\hat{E}$. Training optimizes a composite objective:

$$L_{\text{total}} = L_{\text{denoise}} + \lambda_{\text{energy}} M\hat{S}E(\hat{E} - E_{\text{QM9}})$$

where L_{denoise} is the flow match loss in latent coordinates and $\lambda_{\text{energy}} \in [0, 1]$ controls the trade-off between geometric fidelity and energetic realism. The head’s parameters are updated jointly with the DiT weights; gradients do not flow back into the frozen VAE encoder and decoder, preserving its previously learned geometry-aware latent space.

Fine-tuning starts from the publicly released ADiT checkpoints trained on the mixed periodic + non-periodic corpus, but the energy head is calibrated only on the $\sim 110\text{k}$ non-periodic molecules in QM9 to keep the cost of reference energies manageable. Periodic crystals thus shape the latent manifold through the VAE but do not influence the energy regression term, preventing bias toward periodic bonding patterns. Each model is trained for 30 epochs using AdamW with a learning rate of 1×10^{-4} and evaluated every epoch on held-out QM9 conformers. During inference, the energy head is optional; disabling it yields the same sampling speed as vanilla ADiT, whereas enabling it provides re-ranking of candidates based on predicted energy.

3.4 SANITY CHECKS OF SAMPLED CONFORMERS

We utilize the PoseBusters validation suite, which performs a sequence of chemistry-aware sanity checks on every generated molecule and returns a Boolean pass/fail result for each test. Checks include valency, atom connectivity, bond lengths and angles, aromaticity, steric clash, and InChI convertibility. For a batch of N molecules, PoseBusters reports the fraction of molecules that pass each check. The metrics are averaged over batches to quantify the sample’s quality.

4 RESULTS & DISCUSSIONS

4.1 ENERGY-AWARE ADiT TRAINING

Starting from the public QM9 checkpoint, we fine-tuned the DiT denoiser for 30 epochs under four energy-weight schedules $\lambda_{\text{energy}} \in \{0, 0.01, 0.1, 0.2\}$. Figure 2 shows the evolution of $\mathcal{L}_{\text{total}}$ on the *validation* split; the abscissa counts *mini-batch updates* (iterations) rather than epochs, making all runs directly comparable.

Across all choices of λ_{energy} the loss drops steeply during the first $\sim 10^3$ iterations and reaches a clear plateau by 3×10^3 iterations ($\simeq 25$ epochs). The absence of an upswing afterwards indicates that 30 epochs are sufficient and that no measurable over-fitting occurs, even for the largest weight $\lambda_{\text{energy}} = 0.2$. Runs with higher λ_{energy} begin at a larger total loss, as the energy term initially dominates, but ultimately converge to comparable plateaus, confirming that the model can balance geometric fidelity with energetic realism.

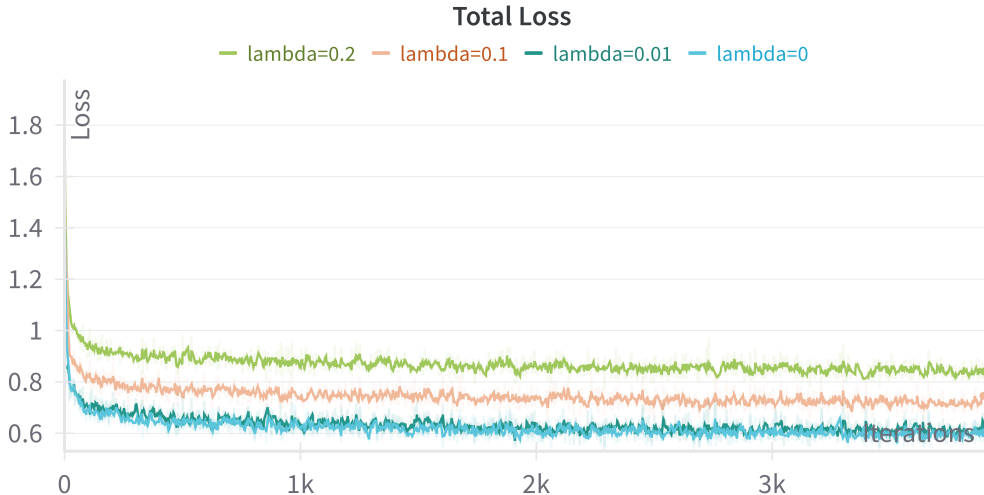


Figure 2: Validation-set loss trajectories for different energy weights λ_{energy} . The horizontal axis represents optimization iterations; 30 training epochs correspond to approximately 3.6×10^3 iterations. All runs plateau well before the iteration budget is exhausted, indicating successful convergence.

4.2 EFFECT OF λ_{ENERGY} ON SAMPLED CONFORMERS

We conduct an ablation study by varying λ_{energy} from 0.0 to 0.2. As λ_{energy} increases, predicted conformers exhibit lower steric clash rates and better internal energy scores, but at the cost of slight diversity loss. Each model was fine-tuned for 30 epochs. Optimal balance is achieved around $\lambda = 0.2$. An increase in λ_{energy} beyond 0.2 results in highly nonphysical structures that do not pass the sanity checks. With $\lambda_{\text{energy}} = 0$ the model behaves like the original ADiT, and increase in λ_{energy} shows an affinity for predicting the lower energy conformer for a given molecule as shown in Figure 3

These trends reveal a classic trade-off between precision and diversity. A small but non-zero energetic prior sharpens the model’s understanding of thermodynamics, suppressing high-energy artifacts without greatly sacrificing coverage of alternative conformations. Excessive weighting, on the other hand, pushes the latent diffusion process into poorly explored regions of latent space, undermining chemical validity. In practice, a fixed value of $\lambda_{\text{energy}} \approx 0.20$ provides a robust operating point; future work could explore curriculum or schedule-based schemes that gradually increase λ_{energy} during fine-tuning, allowing the model to learn structural diversity first before tightening the energetic fidelity.

From Figure 4, although the three rows of molecules appear nearly superimposable at first glance, quantitative analysis reveals that the latent-space energy penalty profoundly modulates their underlying thermodynamics and fine-scale geometry. With no energy penalty ($\lambda_{\text{energy}} = 0$) the diffusion model explores a broad swath of the potential-energy surface, yielding conformers that span high- and low-energy basins; this spread is reflected in the wide distribution of predicted internal energies and in residual geometric distortions that are difficult to spot by eye.

At larger weights ($\lambda_{\text{energy}} \geq 0.1$) the energy head dominates the objective, steering the model toward low-lying energy structures: conformers differ by ≤ 0.4 Å from their ground states, and internal

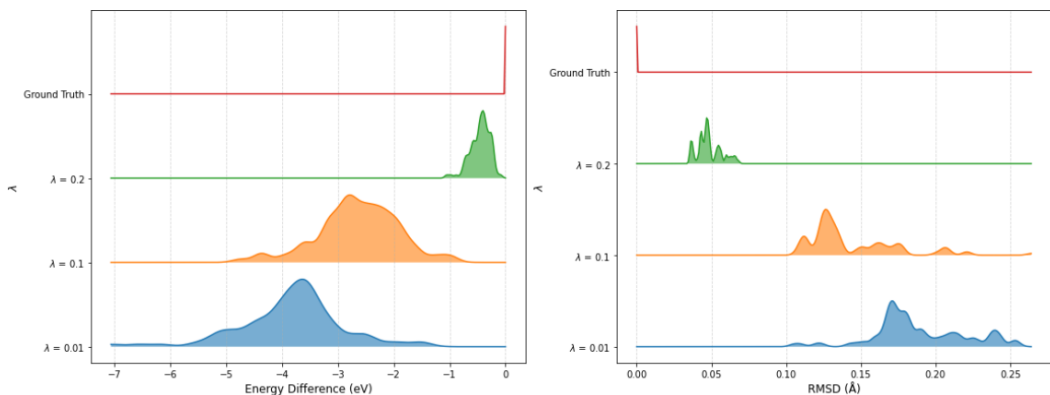


Figure 3: **Left Panel:** Depicts the energy differences of the generated molecules as compared to the ground truth predicted conformer. **Right Panel:** Depicts the RMSD of the generated molecules as compared to the ground truth predicted conformer.

energies are closer to the ground truth predicted values. Thus, what seems like cosmetic variation in 3-D coordinates translates into orders-of-magnitude changes in energetic plausibility, with $\lambda_{\text{energy}} = 0.01$ representing the inflection point where residual geometric artifacts remain most visually evident.

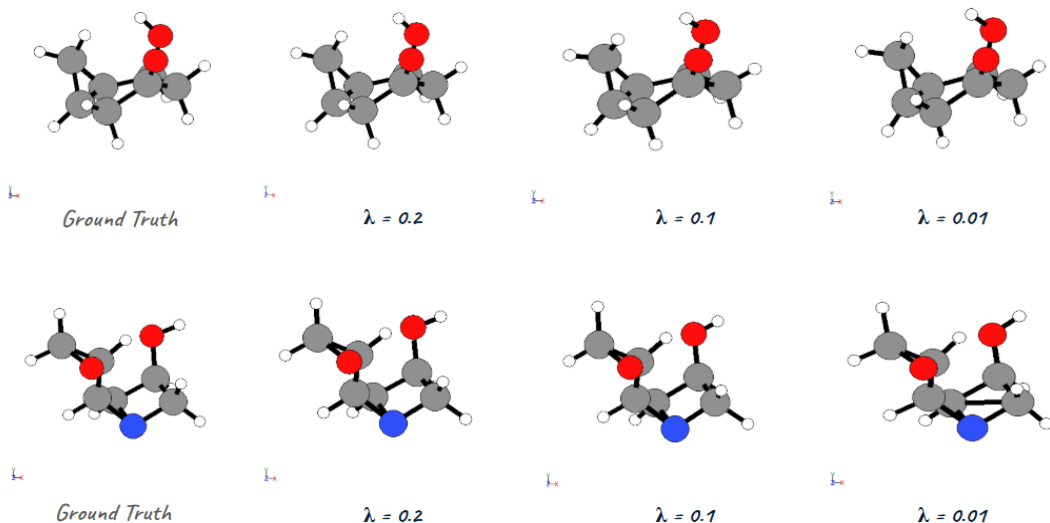


Figure 4: Schematic showing two generated molecules (from 100 sampled molecules) with varying levels of λ_{energy} . Although the structures may appear similar on quick inspection, they differ significantly in energy values and on the Angstrom scale. The differences are more pronounced and visible explicitly in the $\lambda_{\text{energy}} = 0.01$ case

5 CONCLUSIONS

In this work, we revisit the All-Atom Diffusion Transformer (ADiT) as a general-purpose engine for molecular conformer generation and propose an energy-aware fine-tuning scheme that augments the latent diffusion objective with a lightweight, learnable energy head. By coupling the generative loss to reference quantum-mechanical energies, we effectively steered the model toward low-lying regions of the potential-energy surface while preserving the geometric diversity learned during pre-training.

Looking ahead, we see two immediate extensions. (i) Periodic crystals and MOFs: the same two-stage recipe can be applied to periodic lattices by re-introducing lattice vectors in the decoder and conditioning the diffusion model on space-group symmetry. (ii) Force-level supervision: replacing scalar energies with vectorial forces would supply richer gradients, enabling the generator to learn local curvature of the energy landscape

AUTHOR CONTRIBUTIONS

Sudheesh implemented the energy-aware loss, training pipeline, and conducted experiments. Saurabh assisted with model setup, debugging, and testing. Both authors contributed equally to the presentation and report.

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